# 1. hwcost round 2: run records (runs started 20261016T1654Z)

**Kind:** benchmark and analysis runs, descriptive report (observations only; interpretation is in the study report `../../report.pdf`).
**Study:** `research/hwcost_round2_20261016/`, protocol `PROTOCOL.md` frozen at commit `597502b4d` before parts A–D ran; addendum 1 frozen at `4e5994fc9` before runs cm-1 and cm-2.
**Commands:** the protocol's "Commands" sections, verbatim. Every run below exited 0 and completed every requested arm (7 repetitions of about 100 ms each, batches 16, 64, 256, 1024, 4096, 16384, 65536, 262144, 1048576, 4194304); no arm failed or timed out.

## 1.1. Runs

| run | binary | fields | started UTC | finished UTC | checkout HEAD (`commit.txt`) |
|---|---|---|---|---|---|
| isa-base-1 | `target` (default x86-64) | fp64, fp64bf, w1, w2, w3, w4 | 20261016T1654Z | 20261016T1657Z | `597502b4d` |
| isa-native-1 | `target-native` (`-C target-cpu=native`) | fp64, fp64bf, w1, w2, w3, w4 | 20261016T1657Z | 20261016T1700Z | `b427150af` |
| isa-base-2 | `target` (default x86-64) | fp64, fp64bf, w1, w2, w3, w4 | 20261016T1700Z | 20261016T1702Z | `b427150af` |
| isa-native-2 | `target-native` (`-C target-cpu=native`) | fp64, fp64bf, w1, w2, w3, w4 | 20261016T1702Z | 20261016T1705Z | `b427150af` |
| cm-1 | `target` (default x86-64) | fp64, fp64cm | 20261016T1722Z | 20261016T1723Z | `4e5994fc9` |
| cm-2 | `target` (default x86-64) | fp64, fp64cm | 20261016T1723Z | 20261016T1724Z | `4e5994fc9` |

`commit.txt` is the checkout's HEAD when `measure` started. Both part-A binaries were built at 16:54 UTC from the protocol commit's tree; `b427150af` (committed during the runs) changes only claim printing in `compare`, `gpu-check` and `ic-price` and `fit.rs`'s claim helpers: the timed code (`kernel.rs`, `measure.rs`, `mod.rs`, `rho.rs`) is identical at both commits (`git diff --stat 597502b4d b427150af`). cm-1 and cm-2 ran a baseline binary built at `4e5994fc9`.

Compiled target features: baseline runs none probed (x86-64 baseline) native runs sse4.2, avx, avx2, fma, bmi1, bmi2, adx, avx512f, avx512ifma, pclmulqdq, vpclmulqdq.

## 1.2. Host and isolation

- CPU: Intel(R) Xeon(R) Processor @ 2.10GHz (GenuineIntel, signature 6/207/2), 4 logical CPUs, 1 NUMA node; virtualization: cpu_flag = hypervisor, systemd_detect_virt_vm = kvm kernel 6.18.44-fc-v139 rustc 1.97.0 (2d8144b78 2026-07-07).
- Caches (sysfs): L1 Data 48 KiB, L2 Unified 2048 KiB, L3 Unified 266240 KiB.
- Isolation **L0**: shared cloud container, no pinning, no frequency control; arms interleaved, order rotated per repetition. Host class `ECBENV2h55b1cb8b1d10`.
- **No hardware performance counters:** the guest kernel reports `Performance Events: unsupported CPU family 6 model 207 no PMU driver, software events only.` Branch misses, retired instructions and cycles could not be counted. This is missing data, not a measured zero.

## 1.3. Correctness gates (before any timing)

Every run's `stderr.txt` opens with one line per field: the batched walk equals the unbatched reference and stays on the curve, and the timed arithmetic equals the field's (for `fp64bf` and `fp64cm`: the walk is `Fp64`'s point for point, claim F0). All gates passed in all six runs; the unit test `branch_free_fp64_is_bit_for_bit_fp64` compares 10,000 random operations of each variant with `Fp64`.

## 1.4. Part A: baseline vs native build

**Validation claims per run** (median |relative error|; threshold 0.30; C0 threshold 0.05). V1 = calibrated model on test batches B ≥ 4096, V2 = a priori model, V3 = size-extrapolated model.

| claim | isa-base-1 | isa-native-1 | isa-base-2 | isa-native-2 |
|---|---|---|---|---|
| C0 | 0.0028 PASS | 0.001 PASS | 0.007 PASS | 0.0028 PASS |
| V1.fp64 | 0.0417 PASS | 0.0898 PASS | 0.0077 PASS | 0.0706 PASS |
| V1.fp64bf | 0.0228 PASS | 0.0806 PASS | 0.0322 PASS | 0.0496 PASS |
| V1.w1 | 0.0635 PASS | 0.0269 PASS | 0.0211 PASS | 0.0803 PASS |
| V1.w2 | 0.0685 PASS | 0.1199 PASS | 0.0418 PASS | 0.075 PASS |
| V1.w3 | 0.0146 PASS | 0.0219 PASS | 0.0262 PASS | 0.0454 PASS |
| V1.w4 | 0.0625 PASS | 0.0774 PASS | 0.029 PASS | 0.0453 PASS |
| V2.fp64 | 0.5486 FAIL | 0.1451 PASS | 0.5358 FAIL | 0.0816 PASS |
| V2.fp64bf | 0.5298 FAIL | 0.0953 PASS | 0.5505 FAIL | 0.0634 PASS |
| V2.w1 | 0.0404 PASS | 0.0235 PASS | 0.0314 PASS | 0.043 PASS |
| V2.w2 | 0.2032 PASS | 0.2313 PASS | 0.1777 PASS | 0.1511 PASS |
| V2.w3 | 0.1713 PASS | 0.1453 PASS | 0.1378 PASS | 0.1039 PASS |
| V2.w4 | 0.1356 PASS | 0.0843 PASS | 0.1306 PASS | 0.0671 PASS |
| V3.w4 | 0.0938 PASS | 0.058 PASS | 0.0791 PASS | 0.0675 PASS |

**Calibration transfer (I3)**: native op costs plus isa-base-1's fitted residuals (`hwcost fit --residual-from`), median |error| on test batches:

| claim | isa-native-1 | isa-native-2 |
|---|---|---|
| V1.fp64 | 1.5061 FAIL | 1.3843 FAIL |
| V1.fp64bf | 1.3207 FAIL | 1.2942 FAIL |
| V1.w1 | 0.0185 PASS | 0.0178 PASS |
| V1.w2 | 0.0148 PASS | 0.0942 PASS |
| V1.w3 | 0.0374 PASS | 0.0492 PASS |
| V1.w4 | 0.0294 PASS | 0.0219 PASS |

**Per-arm comparison (I1)** (`runs/compare/compare.csv`, 148 arms): $r_k = B_k/A_k$, drift $d = |A_2/A_1 - 1|$, an effect called only when both ratios clear $\max(2d, 0.02)$ on the same side. Counts per arm group:

| group | class | arms |
|---|---|---|
| clock | no detected effect | 2 |
| fp64 | candidate slower | 3 |
| fp64 | no detected effect | 5 |
| fp64bf | candidate slower | 1 |
| fp64bf | no detected effect | 7 |
| kernel | candidate faster | 22 |
| kernel | candidate slower | 1 |
| kernel | no detected effect | 37 |
| mem | candidate faster | 9 |
| mem | candidate slower | 1 |
| mem | no detected effect | 10 |
| scaling | candidate faster | 9 |
| scaling | no detected effect | 9 |
| w1 | candidate slower | 3 |
| w1 | no detected effect | 5 |
| w2 | candidate faster | 1 |
| w2 | no detected effect | 7 |
| w3 | candidate faster | 1 |
| w3 | no detected effect | 7 |
| w4 | candidate faster | 1 |
| w4 | candidate slower | 3 |
| w4 | no detected effect | 4 |

Kernel arms at four batch sizes (ns per rho iteration, medians of 7):

| arm | A1 | B1 | A2 | B2 | r1 | r2 | drift | margin | class |
|---|---|---|---|---|---|---|---|---|---|
| fp64.262144 | 48.3 | 20.7 | 48.1 | 22.2 | 0.429 | 0.462 | 0.004 | 0.02 | candidate faster |
| fp64.4096 | 47.9 | 20.9 | 47.7 | 20.1 | 0.435 | 0.42 | 0.004 | 0.02 | candidate faster |
| fp64.4194304 | 50.9 | 22.1 | 50.6 | 23.3 | 0.435 | 0.46 | 0.006 | 0.02 | candidate faster |
| fp64.64 | 56.5 | 25.6 | 55.7 | 25.6 | 0.454 | 0.46 | 0.015 | 0.029 | candidate faster |
| fp64bf.262144 | 49.3 | 20.3 | 47.9 | 20.7 | 0.412 | 0.433 | 0.028 | 0.057 | candidate faster |
| fp64bf.4096 | 47.2 | 20 | 47.1 | 20.8 | 0.423 | 0.442 | 0.002 | 0.02 | candidate faster |
| fp64bf.4194304 | 52.3 | 22.5 | 51.1 | 22.6 | 0.431 | 0.443 | 0.022 | 0.045 | candidate faster |
| fp64bf.64 | 53.7 | 26 | 55.4 | 26.1 | 0.484 | 0.47 | 0.032 | 0.064 | candidate faster |
| w1.262144 | 33.9 | 31.8 | 31.2 | 32 | 0.937 | 1.028 | 0.081 | 0.163 | no detected effect |
| w1.4096 | 30.1 | 32.3 | 30.9 | 31.4 | 1.073 | 1.018 | 0.024 | 0.047 | no detected effect |
| w1.4194304 | 34.3 | 32.9 | 33.3 | 33.3 | 0.959 | 1 | 0.028 | 0.057 | no detected effect |
| w1.64 | 40.8 | 39.7 | 41 | 40.5 | 0.971 | 0.988 | 0.004 | 0.02 | no detected effect |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| w2.262144 | 79.9 | 85.5 | 86.5 | 80.3 | 1.071 | 0.929 | 0.083 | 0.167 | no detected effect |
| w2.4096 | 79.6 | 84.7 | 78.9 | 75.8 | 1.063 | 0.961 | 0.01 | 0.02 | no detected effect |
| w2.4194304 | 81.4 | 87.9 | 86.4 | 87.2 | 1.08 | 1.009 | 0.062 | 0.123 | no detected effect |
| w2.64 | 108.1 | 96.3 | 101.8 | 96.4 | 0.891 | 0.947 | 0.058 | 0.116 | no detected effect |
| w3.262144 | 145.5 | 144.7 | 146.3 | 143.5 | 0.994 | 0.981 | 0.005 | 0.02 | no detected effect |
| w3.4096 | 143.9 | 139 | 137.1 | 136.7 | 0.966 | 0.997 | 0.047 | 0.094 | no detected effect |
| w3.4194304 | 144.1 | 151.6 | 148.5 | 147.1 | 1.052 | 0.99 | 0.031 | 0.061 | no detected effect |
| w3.64 | 179.6 | 173.5 | 172.1 | 173.4 | 0.966 | 1.008 | 0.042 | 0.083 | no detected effect |
| w4.262144 | 217.3 | 228.1 | 216 | 208.8 | 1.05 | 0.966 | 0.006 | 0.02 | no detected effect |
| w4.4096 | 199.9 | 211.7 | 204.5 | 207.1 | 1.059 | 1.012 | 0.023 | 0.046 | no detected effect |
| w4.4194304 | 218.8 | 225 | 216 | 218.1 | 1.029 | 1.01 | 0.013 | 0.025 | no detected effect |
| w4.64 | 260.6 | 250.3 | 260.8 | 248.8 | 0.96 | 0.954 | 0.001 | 0.02 | candidate faster |

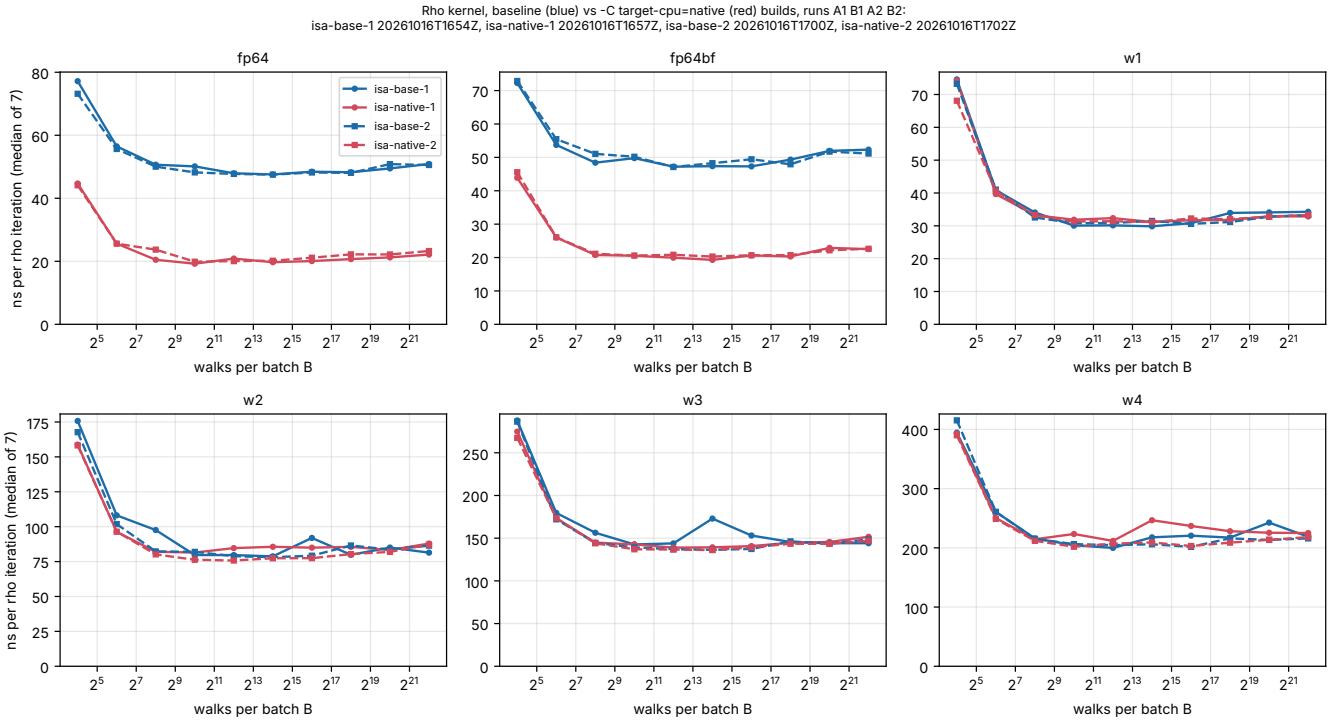



Figure 1: Rho kernel time per field against batch size, runs isa-base-1, isa-native-1, isa-base-2, isa-native-2 (medians of 7 repetitions each).

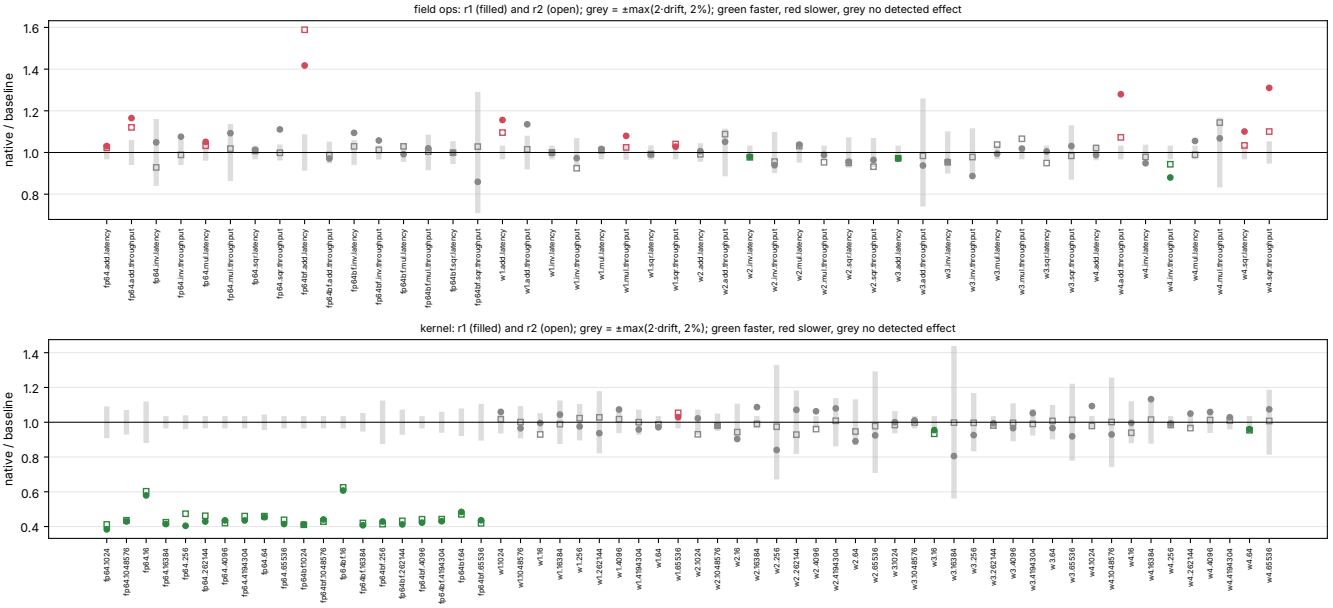



Figure 2: Native / baseline ratio per arm, $r_1$ and $r_2$, with the drift margin, all 148 arms of `compare.csv`.

## 1.5. Part B: `fp64` vs `fp64bf` (baseline runs)

`fp64bf / fp64` kernel ratio at B ≥ 4096 in isa-base-1 and isa-base-2 (F1 cells):

| run | B | fp64 ns | fp64bf ns | fp64bf / fp64 |
|---|---|---|---|---|
| isa-base-1 | 4096 | 47.93 | 47.21 | 0.9849 |
| isa-base-1 | 16384 | 47.57 | 47.36 | 0.9957 |
| isa-base-1 | 65536 | 48.44 | 47.3 | 0.9764 |
| isa-base-1 | 262144 | 48.26 | 49.33 | 1.0221 |
| isa-base-1 | 1048576 | 49.48 | 51.95 | 1.0499 |
| isa-base-1 | 4194304 | 50.88 | 52.32 | 1.0283 |
| isa-base-2 | 4096 | 47.73 | 47.11 | 0.987 |
| isa-base-2 | 16384 | 47.51 | 48.26 | 1.0158 |
| isa-base-2 | 65536 | 48.18 | 49.43 | 1.0259 |
| isa-base-2 | 262144 | 48.07 | 47.93 | 0.9971 |
| isa-base-2 | 1048576 | 50.85 | 51.71 | 1.017 |
| isa-base-2 | 4194304 | 50.56 | 51.15 | 1.0116 |

**F1** (worst fp64bf/fp64 kernel ratio at B ≥ 4096 in the baseline runs (12 cells)): 1.0499, threshold 0.85, FAIL.

**F3, disassembly** (`runs/f3/`, `f3_disasm.sh`; `objdump -d` of each `#[inline(never)]` kernel function; binaries in `runs/f3/binaries.txt`):

| build | function | instructions | cond. jumps | cmov | vector-register | mulx |
|---|---|---|---|---|---|---|
| baseline | `step_fp64` | 303 | 32 | 4 | 0 | 0 |
| baseline | `step_fp64bf` | 303 | 32 | 4 | 0 | 0 |
| baseline | `step_fp64cm` | 285 | 20 | 12 | 0 | 0 |
| native | `step_fp64` | 337 | 22 | 14 | 38 | 12 |
| native | `step_fp64bf` | 337 | 22 | 14 | 38 | 12 |

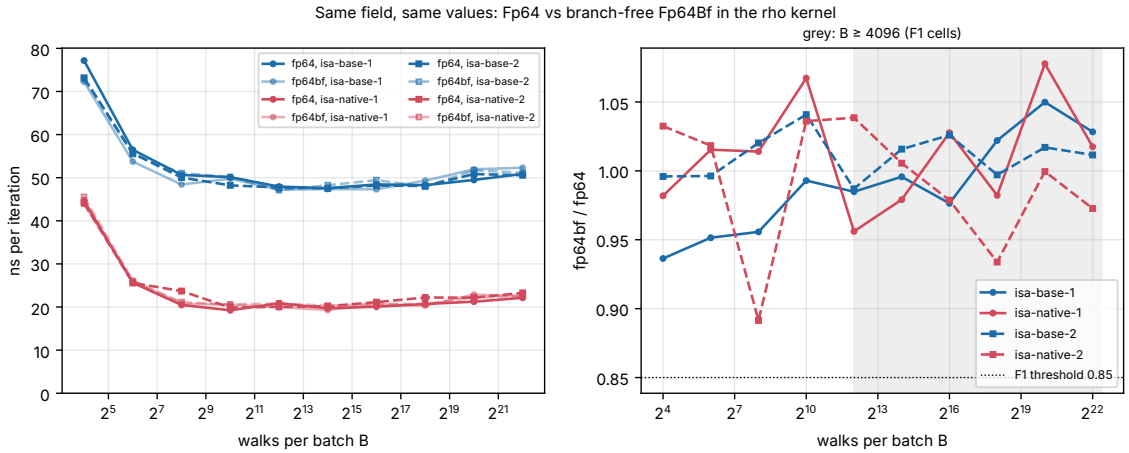



Figure 3: `fp64` and `fp64bf` kernels in all four part-A runs, and their ratio; grey band = F1 cells.

## 1.6. Addendum 1: `fp64` vs `fp64cm` (runs cm-1, cm-2)

| run | B | fp64 ns | fp64cm ns | fp64cm / fp64 |
|---|---|---|---|---|
| cm-1 | 16 | 76.79 | 45.07 | 0.5869 |
| cm-1 | 64 | 55.35 | 26.07 | 0.471 |
| cm-1 | 256 | 49.71 | 21.78 | 0.4383 |
| cm-1 | 1024 | 48.18 | 20.85 | 0.4327 |
| cm-1 | 4096 | 48.41 | 20.28 | 0.419 |
| cm-1 | 16384 | 46.66 | 20.36 | 0.4364 |
| cm-1 | 65536 | 48.08 | 21.27 | 0.4424 |
| cm-1 | 262144 | 48.97 | 21.25 | 0.434 |
| cm-1 | 1048576 | 49.76 | 22.16 | 0.4452 |

| | | | | |
|---|---|---|---|---|
| cm-1 | 4194304 | 49.21 | 24.59 | 0.4998 |
| cm-2 | 16 | 72.2 | 44.17 | 0.6117 |
| cm-2 | 64 | 60.03 | 26.44 | 0.4404 |
| cm-2 | 256 | 49.27 | 21.78 | 0.4421 |
| cm-2 | 1024 | 48.33 | 21.66 | 0.4482 |
| cm-2 | 4096 | 50.96 | 21.71 | 0.4261 |
| cm-2 | 16384 | 51.27 | 21.2 | 0.4135 |
| cm-2 | 65536 | 52.7 | 21.34 | 0.4048 |
| cm-2 | 262144 | 48.48 | 20.25 | 0.4176 |
| cm-2 | 1048576 | 49.56 | 21.3 | 0.4298 |
| cm-2 | 4194304 | 50.07 | 23.08 | 0.4609 |

**F1prime** (worst fp64cm/fp64 kernel ratio at B ≥ 4096 over 2 runs (12 cells)): 0.4998, threshold 0.85, PASS.

| claim | cm-1 | cm-2 |
|---|---|---|
| C0 | 0.009 PASS | 0.0385 PASS |
| V1.fp64 | 0.0149 PASS | 0.031 PASS |
| V1.fp64cm | 0.0587 PASS | 0.0679 PASS |
| V2.fp64 | 0.5407 FAIL | 0.554 FAIL |
| V2.fp64cm | 0.0791 PASS | 0.0425 PASS |

Fitted residual per iteration (model C): cm-1 fp64 27.2 ns, fp64cm −2 ns; cm-2 fp64 27 ns, fp64cm −2 ns.

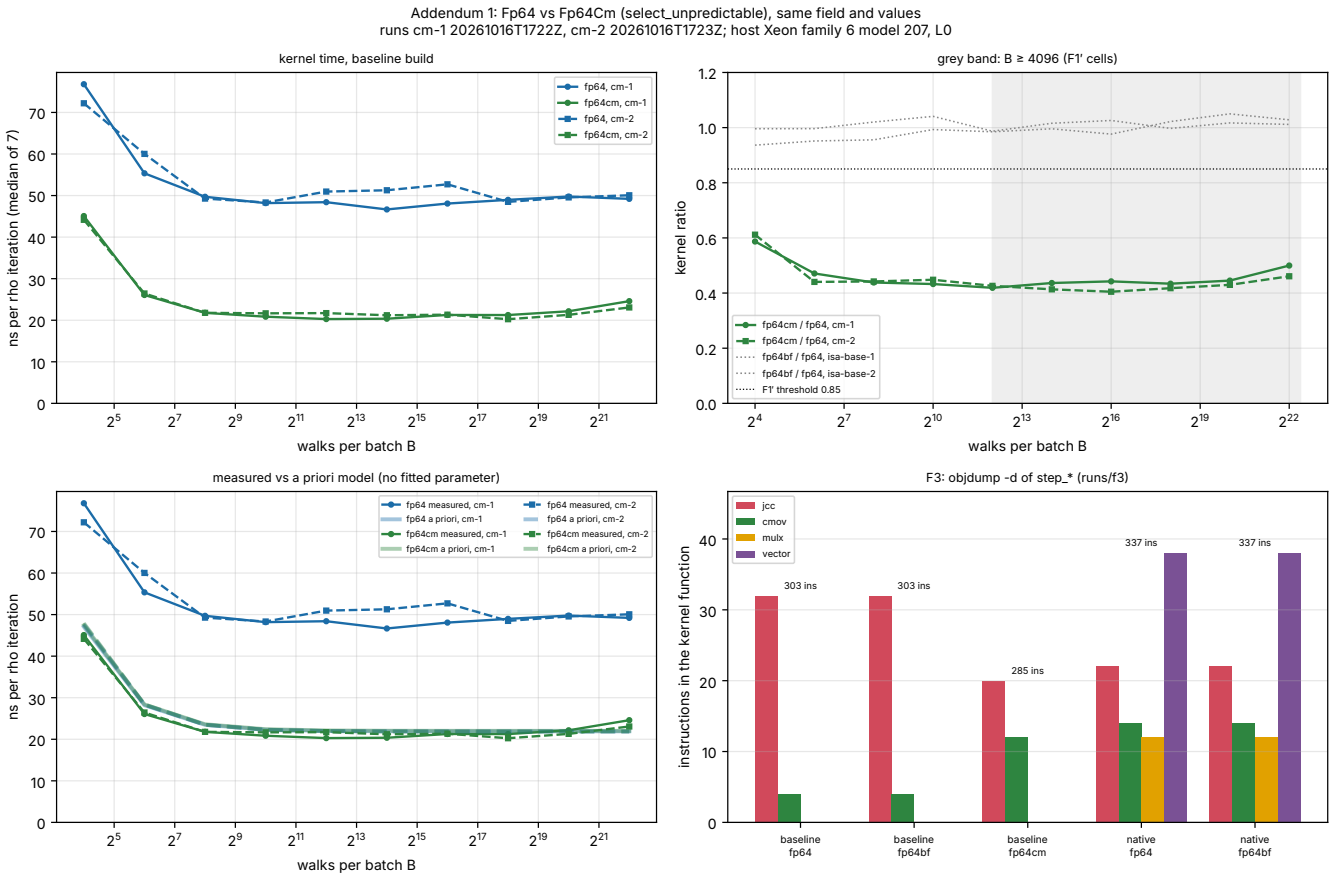



Figure 4: Runs cm-1 and cm-2: kernel time, `fp64cm / fp64` ratio (part A's `fp64bf / fp64` dotted for reference), measured against the a priori model, and the F3 instruction counts.

## 1.7. Part C: GPU profile against 46 ECC2K-130 builds

Input: `ecc2k130/benchmarks/roofline/calibration.json` (recorded rates; no GPU ran in this study). Output `runs/gpu/` (per-build rows in `gpu_check.csv`).

| id | statement | value | threshold | verdict |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| G1 | a priori, all 46 builds, median \|error\| | 0.1792 | 0.3 | PASS |
| G1.eta>1 | builds measured above the a priori ceiling (η > 1) | 0 | — | REPORT |
| G2 | η from each GPU's base build, 44 other builds, median \|error\| | 0.0145 | 0.3 | PASS |
| G2.10 | the same, against 10% | 0.0145 | 0.1 | PASS |
| G3 | η from the RTX PRO 6000 base build carried to 25 B200 builds, median \|error\| | 0.195 | 0.3 | PASS |
| G1.shared | shared-int pipes: a priori, median \|error\| | 0.1237 | — | REPORT |
| G1.shared.eta>1 | shared-int pipes: builds with η > 1 | 0 | — | REPORT |
| G2.shared | shared-int pipes: η per GPU, median \|error\| | 0.0192 | — | REPORT |
| G3.shared | shared-int pipes: η carried across GPUs, median \|error\| | 0.1072 | — | REPORT |

Cost-arithmetic check (`summary.txt`): `expected ECC2K-130 work 2^60.9 at 14.63753 B/s on rtx-pro-6000 ($3.031/h): 40827.8 h, $123749`
`expected ECC2K-130 work 2^60.9 at 15.115792 B/s on rtx-pro-6000 ($3.031/h): 39536.1 h, $119834`

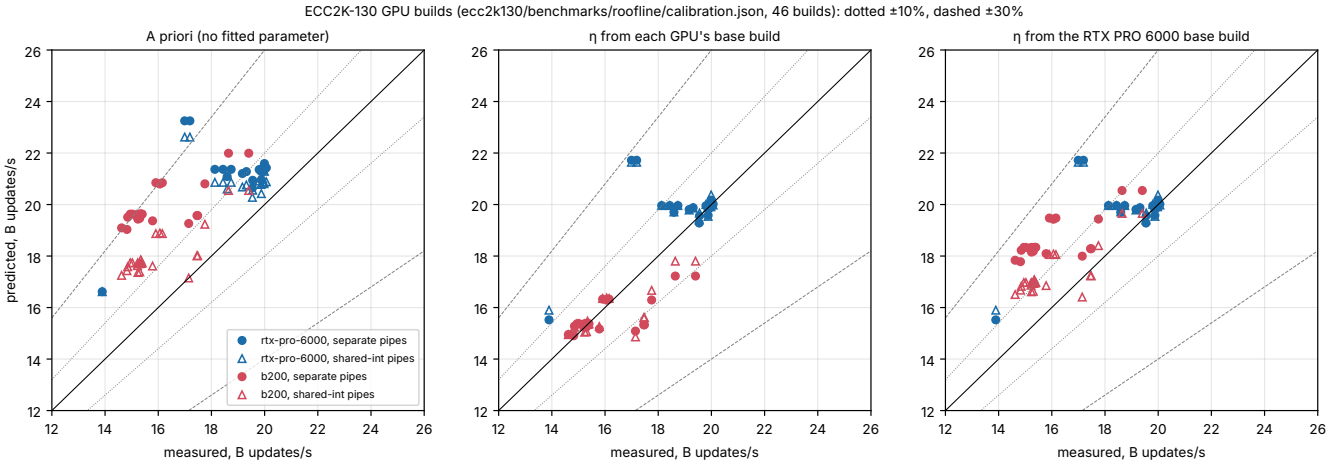



Figure 5: Predicted against measured update rate, 46 builds, both pipe models; the three panels are G1, G2, G3.

## 1.8. Part D: IC and rho in wall time (frozen campaign records)

Input: `research/ic_frontier_campaign_20261015` Phase 1 sessions and §5a band files (32 counted, verified runs per arm and curve; host Xeon 2.30 GHz, 4 vCPU, L0). No solve ran in this study. Dollars at $0.1785/h ÷ 4 per vCPU-hour (illustrative). The `rank` column applies the §5a rule (wall ratio and all three band ratios on one side of 1) per measured curve; it was added to `ic-price` after the analysis and changes no other value.

**koblitz**: ic-compact vs rho-frob.

| curve | log2 r | IC ms | rho ms | wall IC/rho | band w=0 | w=0.1 | w=1 | in band | w* | rank |
|---|---|---|---|---|---|---|---|---|---|---|
| K13 | 11 | 0.15 | 3.02 | 0.049 | 6.87 | 9.33 | 27.1 | false | −0.254 | unranked |
| K17 | 16 | 0.25 | 3.07 | 0.081 | 3.8 | 4.91 | 11.85 | false | −0.28 | unranked |
| K19 | 18 | 0.32 | 3.25 | 0.099 | 4.88 | 7.46 | 23.6 | false | −0.165 | unranked |
| K23 | 22 | 1.06 | 3.57 | 0.296 | 5.48 | 7.39 | 18.45 | false | −0.226 | unranked |
| K37 | 27.8 | 4.35 | 4.15 | 1.049 | 5.2 | 8.31 | 23.29 | false | −0.111 | IC > rho |
| K43 | 32.1 | 22.77 | 3.03 | 7.504 | 7.35 | 10.17 | 23.14 | true | 0.005 | IC > rho |
| K47 | 36.6 | 141.39 | 13.44 | 10.52 | 9.92 | 12.7 | 25.28 | true | 0.02 | IC > rho |
| K41 | 39 | 498.9 | 25.69 | 19.424 | 15.02 | 17.79 | 30.3 | true | 0.169 | IC > rho |

D1.in_band: 3 (curves (of 8) whose measured IC/rho wall ratio lies inside the band's ratio range)
D1.w_star_median: −0.1379 (median implied weight w* (GAE per unpriced native operation))
D1.rank.IC>rho: 4 (curves (of 8) ranked IC > rho (§5a: wall ratio and all three band ratios agree))
D1.rank.IC<rho: 0 (curves (of 8) ranked IC < rho (§5a: wall ratio and all three band ratios agree))
D1.rank.unranked: 4 (curves (of 8) ranked unranked (§5a: wall ratio and all three band ratios agree))

Fitted exponents $\log_2 y = a + \alpha \log_2 r$ over the tier: ic wall_ms α = 0.602 (4 pts); ic P(w=0) α = 0.586 (4 pts); ic P(w=0.1) α = 0.551 (4 pts); ic P(w=1) α = 0.497 (4 pts); rho wall_ms α = 0.255 (4 pts); rho P(w=0) α = 0.457 (4 pts); rho P(w=0.1) α = 0.459 (4 pts); rho P(w=1) α = 0.466 (4 pts).

Pre-registered projection (wall-time exponents; extrapolated, not measured):

| log2 r | IC core-yr | rho core-yr | wall ratio | band w=0 | w=1 | rank |
|---|---|---|---|---|---|---|
| 48 | 5.880844e-7 | 3.129375e-9 | 187.9 | 30.5 | 34.3 | IC > rho |
| 64 | 4.654268e-4 | 5.311050e-8 | 8763.4 | 126.5 | 48.3 | IC > rho |
| 80 | 3.683521e-1 | 9.013703e-7 | 408657.9 | 524.7 | 68 | IC > rho |
| 96 | 2.915244e2 | 1.529770e-5 | 19056751.1 | 2176.5 | 95.7 | IC > rho |
| 112 | 2.307208e5 | 2.596265e-4 | 888664447.4 | 9028.9 | 134.8 | IC > rho |
| 128 | 1.825991e8 | 4.406277e-3 | 41440668188.7 | 37455.9 | 189.8 | IC > rho |

Post hoc projection (`projections_gae.csv`, addendum 1): fitted $P(w)$ exponents at the matched rho's ns per GAE at its largest measured size, log2 r = 128:

| w | ns/GAE | IC core-yr | rho core-yr | IC $ | IC/rho |
|---|---|---|---|---|---|
| 0 | 256.460 | 5.282051e7 | 1.410205e3 | 2.066247e10 | 37455.9 |
| 0.1 | 233.234 | 7.167888e6 | 1.547560e3 | 2.803954e9 | 4631.7 |
| 1 | 128.499 | 4.526957e5 | 2.385653e3 | 1.770867e8 | 189.8 |

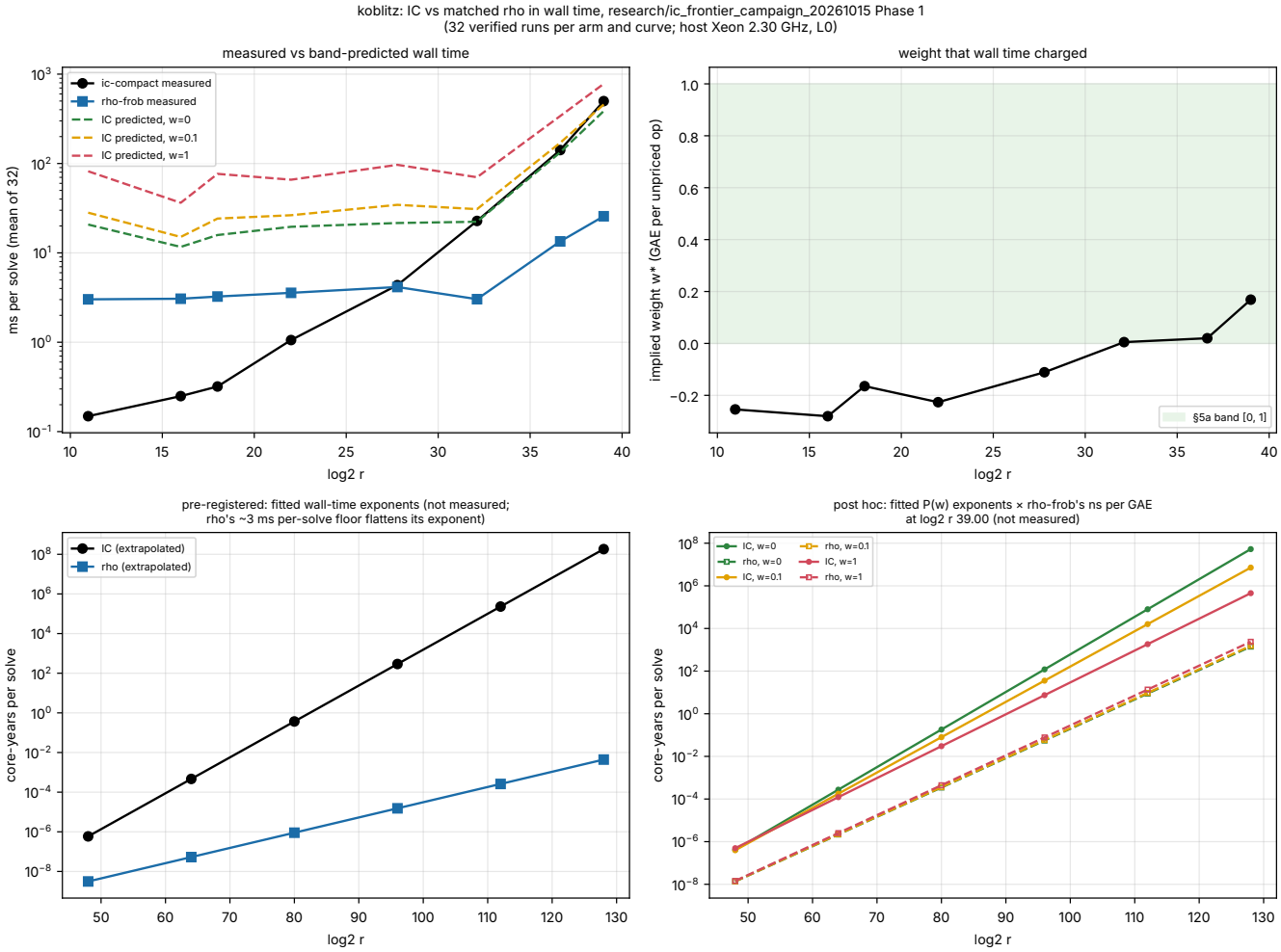



Figure 6: koblitz: measured and band-predicted wall time per solve (mean of 32), implied weight $w^*$, and the two projections (extrapolations, not measurements).

**prime**: ic-pair vs rho-neg.

| curve | log2 r | IC ms | rho ms | wall IC/rho | band w=0 | w=0.1 | w=1 | in band | w* | rank |
|---|---|---|---|---|---|---|---|---|---|---|
| P20 | 20 | 2.44 | 3.71 | 0.658 | 9.82 | 10.46 | 13.51 | false | −0.615 | unranked |
| P24 | 24 | 8.89 | 4.44 | 2.003 | 12.09 | 12.23 | 12.89 | false | −0.929 | IC > rho |
| P28 | 28 | 59.49 | 6.66 | 8.93 | 18.59 | 18.64 | 18.86 | false | −0.966 | IC > rho |
| P32 | 32 | 544.24 | 11.68 | 46.604 | 40.87 | 40.91 | 41.08 | false | −1.085 | IC > rho |
| P36 | 36 | 4913.35 | 51.48 | 95.435 | 58.15 | 58.03 | 57.48 | false | −0.967 | IC > rho |
| P40 | 40 | 63517.38 | 171.71 | 369.907 | 95.22 | 101.8 | 131.41 | false | −1.359 | IC > rho |

D1.in_band: 0 (curves (of 6) whose measured IC/rho wall ratio lies inside the band's ratio range)
D1.w_star_median: −0.9668 (median implied weight w* (GAE per unpriced native operation))
D1.rank.IC>rho: 5 (curves (of 6) ranked IC > rho (§5a: wall ratio and all three band ratios agree))
D1.rank.IC<rho: 0 (curves (of 6) ranked IC < rho (§5a: wall ratio and all three band ratios agree))
D1.rank.unranked: 1 (curves (of 6) ranked unranked (§5a: wall ratio and all three band ratios agree))

Fitted exponents $\log_2 y = a + \alpha \log_2 r$ over the tier: ic wall_ms α = 0.799 (5 pts); ic P(w=0) α = 0.659 (5 pts); ic P(w=0.1) α = 0.663 (5 pts); ic P(w=1) α = 0.68 (5 pts); rho wall_ms α = 0.337 (5 pts); rho P(w=0) α = 0.469 (5 pts); rho P(w=0.1) α = 0.47 (5 pts); rho P(w=1) α = 0.472 (5 pts).

Pre-registered projection (wall-time exponents; extrapolated, not measured):

| log2 r | IC core-yr | rho core-yr | wall ratio | band w=0 | w=1 | rank |
|---|---|---|---|---|---|---|
| 48 | 1.385444e-4 | 2.649696e-8 | 5228.7 | 286 | 376.5 | IC > rho |
| 64 | 9.808659e-1 | 1.118364e-6 | 877054.2 | 2352.4 | 3766.4 | IC > rho |
| 80 | 6.944329e3 | 4.720309e-5 | 147115995.2 | 19348.1 | 37680.8 | IC > rho |
| 96 | 4.916442e7 | 1.992313e-3 | 24677057813.7 | 159132.5 | 376977.5 | IC > rho |
| 112 | 3.480740e11 | 8.409007e-2 | 4139299614988.4 | 1308817.6 | 3771466.7 | IC > rho |
| 128 | 2.464293e15 | 3.549212e0 | 694321074740973.3 | 10764634.7 | 37731591.7 | IC > rho |

Post hoc projection (`projections_gae.csv`, addendum 1): fitted $P(w)$ exponents at the matched rho's ns per GAE at its largest measured size, log2 r = 128:

| w | ns/GAE | IC core-yr | rho core-yr | IC $ | IC/rho |
|---|---|---|---|---|---|
| 0 | 186.197 | 1.415294e11 | 1.314763e4 | 5.536387e13 | 10764634.7 |
| 0.1 | 169.277 | 1.920460e11 | 1.362491e4 | 7.512507e13 | 14095215.6 |
| 1 | 93.120 | 6.053500e11 | 1.604358e4 | 2.368025e14 | 37731591.7 |

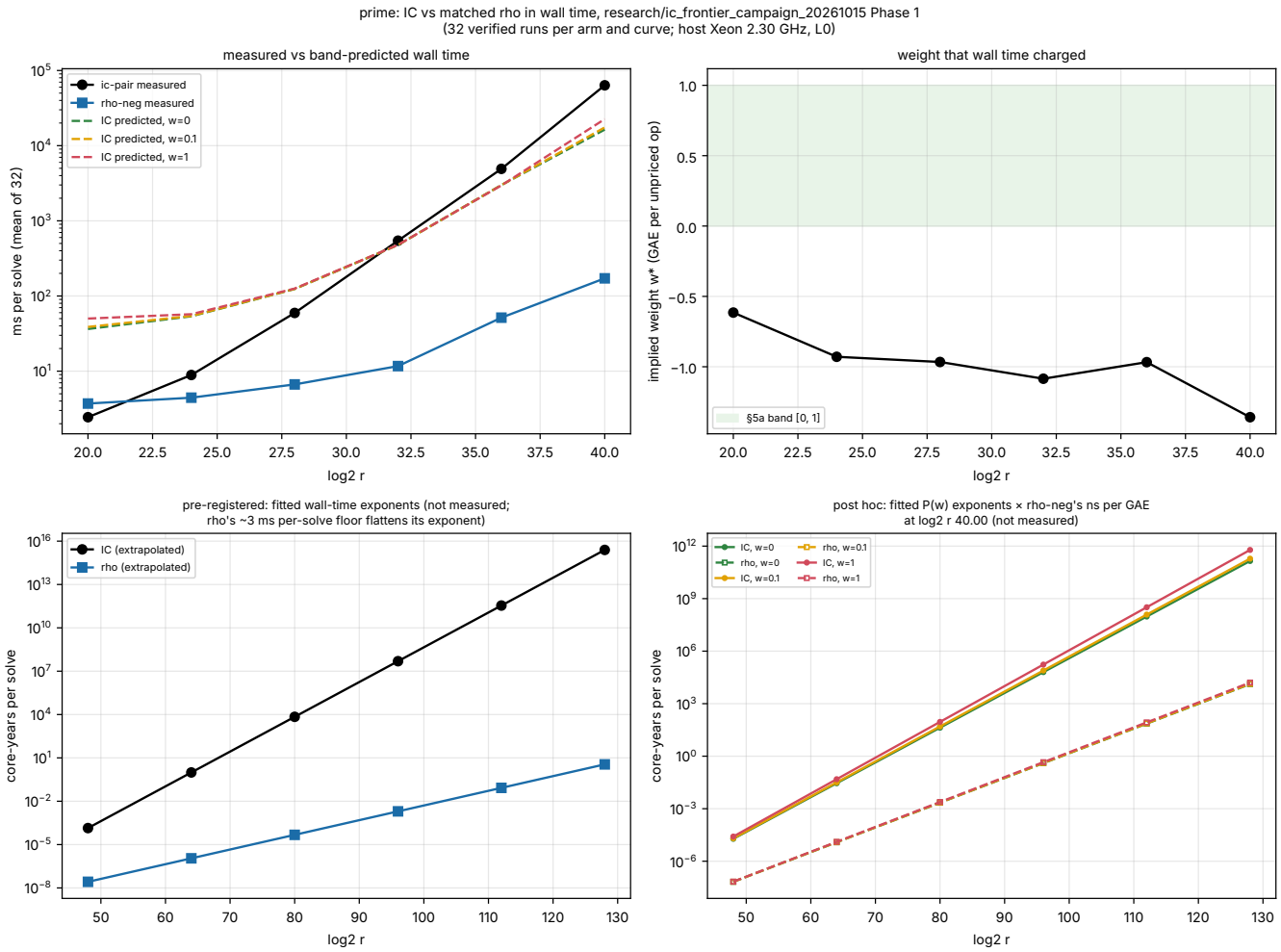



Figure 7: prime: measured and band-predicted wall time per solve (mean of 32), implied weight $w^*$, and the two projections (extrapolations, not measurements).

### 1.9. Deviations and missing data

- No PMU: branch-miss counts are missing; part B and addendum 1 test the branch hypothesis by substitution instead.
- F3 found `step_fp64` and `step_fp64bf` identical in the baseline binary, so F1′s two arms ran the same instructions; addendum 1 (frozen before cm-1 and cm-2) added `fp64cm`.
- Part D's rho wall time carries a per-solve floor of about 3 ms at log2 r ≤ 32 (the `rho ms` column); the post hoc GAE projection was added for that reason and is labelled post hoc.
- `run,batch,…` headers of `cm-pair/pairs.csv` name the compared fields (`fp64cm_ns`); the values were unchanged by that fix.

### 1.10. Raw records

`runs/isa-*/` and `runs/cm-*/`: `measurements.json`, `samples.csv`, `medians.csv`, `validation.{json,csv}`, `profile.json`, `fit.txt`, `stderr.txt`, timestamps, `commit.txt`; `runs/isa-native-*/transfer/`; `runs/compare/`; `runs/cm-pair/`; `runs/f3/`; `runs/gpu/`; `runs/ic-koblitz/`, `runs/ic-prime/`. Graph sources: `../../plot.py` (layout only).